Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the test accuracy during training.

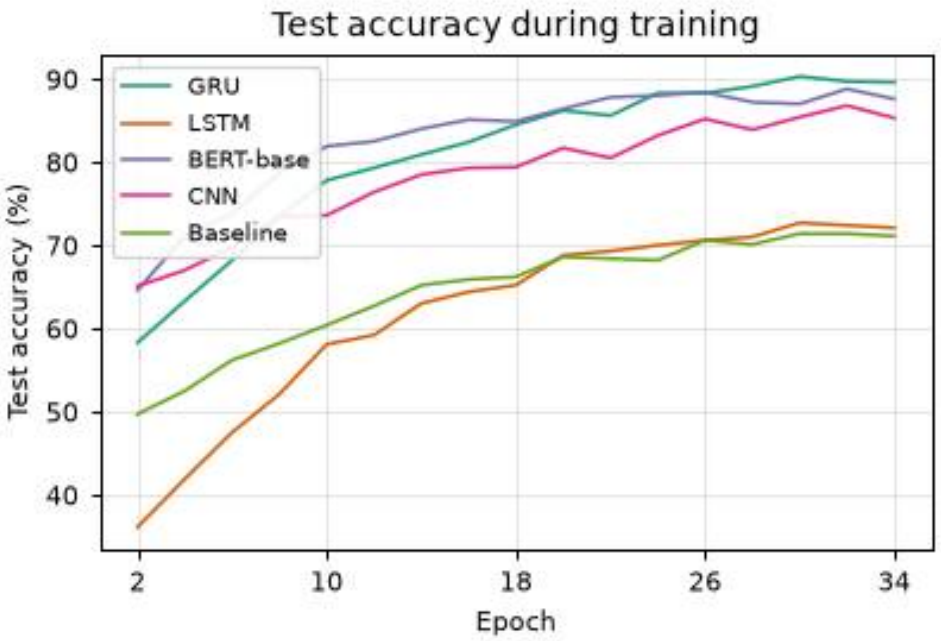


Figure 1: Test accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 34, GRU is 4.3 points better than CNN. As shown in Figure 1, at epoch 14, GRU achieves 83.9% test accuracy.

Figure 2 shows the f1 score during training.

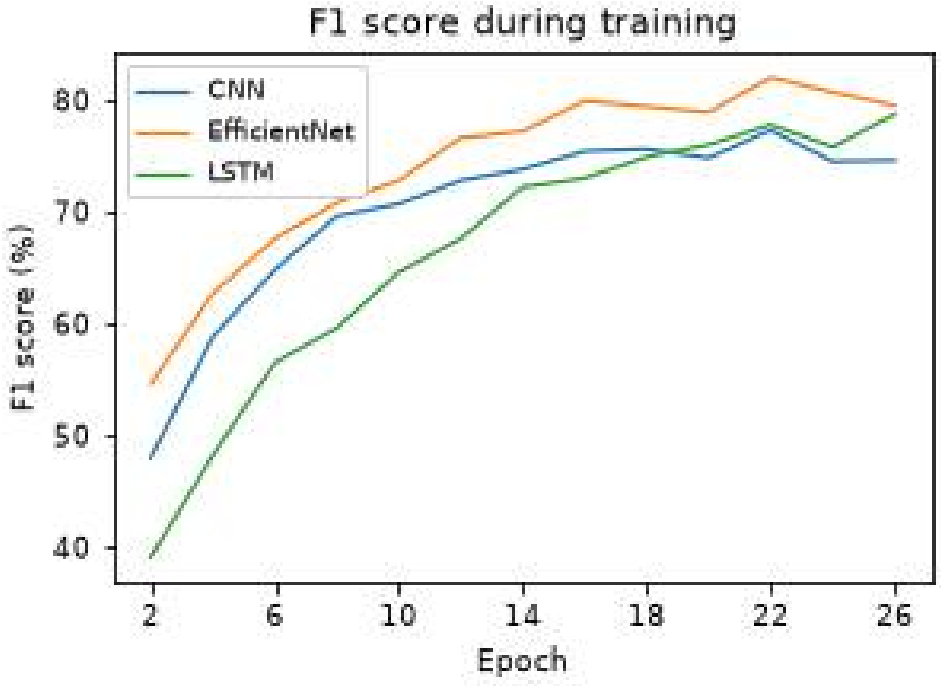


Figure 2: F1 score during training for the models in this experiment.

As shown in Figure 2, at epoch 8, CNN achieves 46.0% F1 score. LSTM attains a F1 score of 74.9% at epoch 24 (Figure 2). At epoch 18, CNN achieves 74.7% F1 score.

Figure 3 shows the test accuracy by dataset.

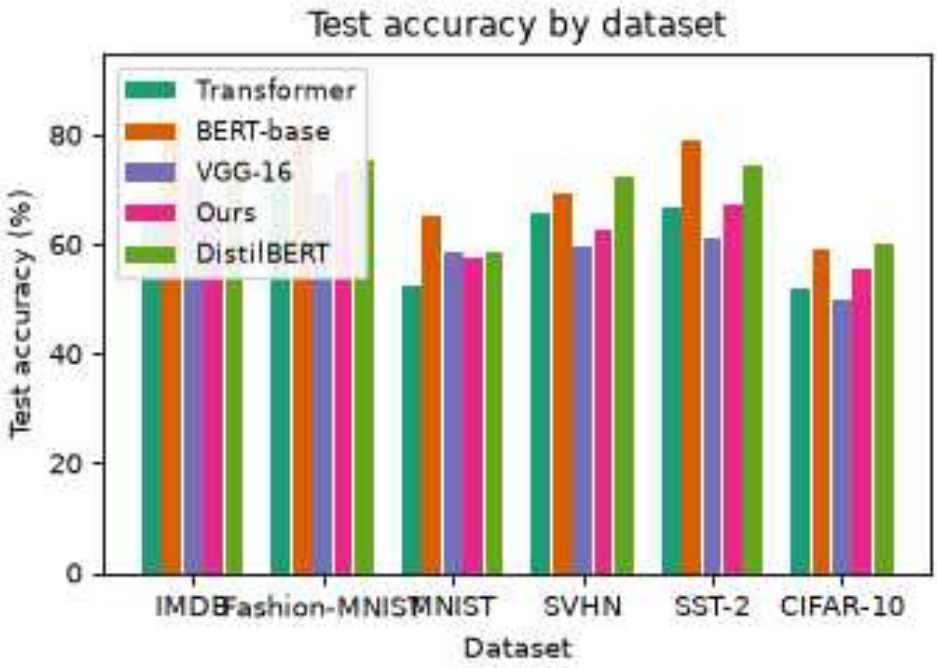


Figure 3: Test accuracy by dataset for the models in this experiment.

DistilBERT achieves 63.0% test accuracy on MNIST (Figure 3). As shown in Figure 3, Ours reaches 62.2% test accuracy on MNIST. BERT-base outperforms Transformer by 59.8 percentage points on MNIST (Figure 3).

Figure 4 shows the f1 score during training.

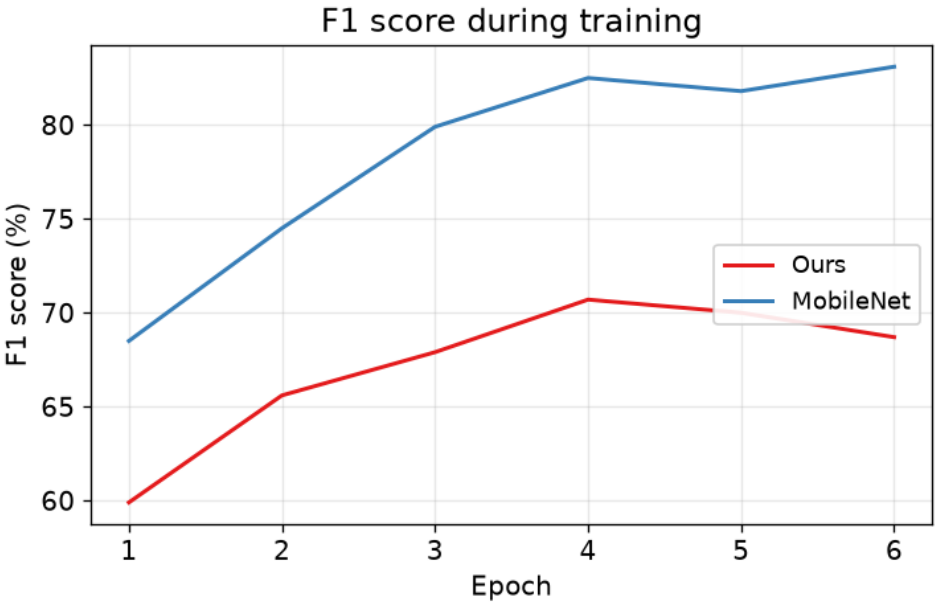


Figure 4: F1 score during training for the models in this experiment.

As shown in Figure 4, MobileNet outperforms Ours by 14.6 percentage points at epoch 3. Ours attains a F1 score of over 65% at epoch 6 (Figure 4).

3 Discussion

On STL-10, XGBoost obtains an accuracy of 89.5%. Overall, the results are in line with what we expected from the literature.